Zagazig University
Faculty of Computers & Informatics
Mid Term Exam Academic Year: 2023-2024

Department of Computer Science
Study Year/Level: First Year
Course Title: Discrete Structures
Course Code: CS103

Model

Examination Date: 20/04/2024
Examination Starts: Morning
Allowed Examination Time: 1 Hour
Total Marks: 20

Student Name: **Student ID:**

- **Authorized Materials:**
- **Calculators allowed. No other additional materials or aides are allowed.**
- **Instructions to Invigilators:**
- **The exam paper may be taken out of the examination room after halftime.**
- **Instructions to Students:**
- **If you feel ill, let us know immediately.**
- **Answer All of the following questions.**
- **If you get stuck on a question, move on, and come back later if you have time.**
- **Remember to time yourself.**

Enjoy!

1. **(2 points)** Determine the cardinality of each of the following sets.
   a) $\{1, \phi, a, 2, z, \{1, 2\}, Z, \{a, z\}, \{a, 1, 2, z\}\}$
   b) $\{x \mid x \text{ is an odd integer and } 1 \le x \le 22\}$
   c) $P(\{1,2,3,4\})$
   d) $\{1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5\}$
2. **(2 points)** Determine whether each of the following statements is true or false.
   a) $\phi \in \phi$
   b) $\phi \subseteq \phi$
   c) $\phi \subset \phi$
   d) $\phi \subseteq \{\phi\}$
3. **(2 points)** Write an **explicit** formula for each of the following sequences:
   a) 0, 2, 0, 2, ...
   b) 1, 1/5, 1/9, 1/13, ...
4. **(2 points)** Suppose R and S are relations on $\{a, b, c, d\}$, where $R = \{(a, b),(a, d),(b, c),(c, c),(d, a)\}$ and $S = \{(a, c),(b, d),(d, a)\}$
   a) Find R^2
   b) Find S^3
5. **(2 points)** Suppose $\mathbf{M}_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, determine if R is:
   a) Reflexive
   b) Symmetric
   c) Antisymmetric
   d) Transitive

6. **(5 points)** Given the set of numbers $S = \{2, 3, 4, 6, 12, 30, 60\}$. The relation R on S is defined as follows: $(x, y) \in R$ if and only if x is a divisor of y.

   a) **(1 pts)** Explain that this relation is a partial order.

   b) **(2 pts)** Draw the Hasse diagram of R.

   c) **(2 pts)** Find the Unit element/ Zero element, and the minimal/maximal elements of R, if any.

      a) Unit element:

      b) Zero element:

      c) Minimal elements:

      d) Maximal element:

7. **(5 points)** Use Dijkstra's algorithm to find the shortest path from source vertex 0 to all other vertices. Then, answer the following questions:

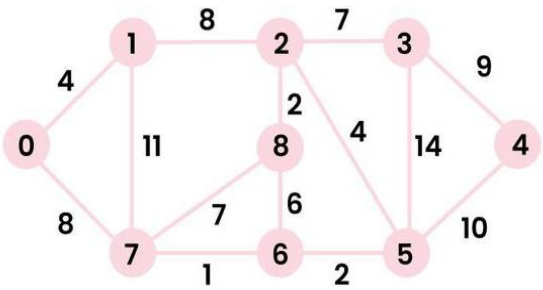
   a) **1 pts)** The shortest path to visit 4 from 0 is ..... with cost.....

   b) **1 pts)** The shortest path to visit 8 from 0 is ..... with cost.....

   c) **1 pts)** The shortest path to visit 5 from 0 is ..... with cost.....

   d) **1 pts)** The shortest path to visit 3 from 0 is ..... with cost.....

   e) **1 pts)** The shortest path to visit 7 from 0 is ..... with cost.....



Good Luck *Dr. Abduallah*